

Robustness: Noisy Signals, Missing PPG, ECG Artifacts, Delayed Measurements and Sensor Failure

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Research prototype. Robustness is evaluated on the synthetic replay set (2 held out patients) with three models trained on the synthetic cohort: concatenation (the v1 architecture, zero fill for missing streams) and two mask aware fusion variants (gated and static weighted). Point estimates, no confidence intervals, because 2 patients cannot support them.

1 Protocol

Waveform perturbations are applied to the cached 30 second replay segments and re embedded through the real foundation model encoders (ECG FM at 500 Hz, PaPaGei at 125 Hz), so the measured robustness includes the encoders and not just the downstream model. Feature level perturbations act on the window vectors directly. The metric is the 6 hour AUROC on the replay windows, scored the way the notebook scores them (step 1, patient level sequences).

The quality gates are deliberately bypassed: flatline and saturation windows would normally be rejected before reaching the model, so feeding them through anyway tests the stronger condition, whether the pipeline degrades even when the gates do not catch the corruption.

2 Results

Every perturbation lands within 0.03 of clean. The only directional effect is the delay family, which degrades monotonically but mildly (concat 0.929 to 0.920 at 15 minutes of staleness). Missing PPG, sensor failure, vitals noise and all ECG artifacts leave the score unchanged.

3 What each test shows

Noisy signals. The ECG FM embeddings absorb white noise down to SNR 5 dB with no measurable change in the risk score, and PaPaGei embeddings likewise at 5 dB. Vitals noise up to twice the per column standard deviation changes nothing, which is consistent with the temporal model averaging 6 hours of history.

Missing PPG and sensor failure. Removing the PPG stream, whether from the start or mid stay, leaves the score unchanged for all three models. The mask aware variants route the absent block to their learned missing embedding; concatenation falls back to zero fill, and on synthetic data the two behaviors are indistinguishable because the PPG block carries little risk signal there. The mask comparison matters where PPG genuinely matters, which is the real data regime covered by the SDDb evaluation.

ECG artifacts. Baseline wander, mains interference, half of each segment flatlined, and hard saturation all leave the score within noise of clean. This is the stronger result: the corruption was fed through the encoders instead of being caught by the quality gates, and the model still held.

Table 1: 6 h AUROC under perturbation (clean baseline 0.929 to 0.945).

Perturbation	concat	gate	weight
Clean	0.929	0.945	0.943
Missing PPG	0.937	0.952	0.947
Sensor failure (PPG dies mid stay)	0.929	0.945	0.943
Delay 5 min	0.926	0.943	0.941
Delay 10 min	0.923	0.942	0.940
Delay 15 min	0.920	0.941	0.938
Vitals noise 0.5x std	0.929	0.945	0.943
Vitals noise 1.0x std	0.929	0.945	0.943
Vitals noise 2.0x std	0.929	0.951	0.943
ECG noise 20 dB	0.929	0.945	0.943
ECG noise 10 dB	0.931	0.948	0.944
ECG noise 5 dB	0.934	0.949	0.945
ECG baseline wander	0.929	0.945	0.944
ECG 50 Hz mains	0.929	0.944	0.943
ECG flatline halves	0.928	0.940	0.952
ECG saturation	0.929	0.946	0.943
PPG noise 5 dB	0.958	0.949	0.963

Delayed measurements. The only family with a directional effect. Shifting the whole monitor vector back by 5, 10 and 15 minutes degrades the concat score from 0.929 to 0.920, a real but mild monotonic loss, and the mask aware variants show the same slope (0.945 to 0.938). A 15 minute staleness budget is therefore tolerable in this regime.

4 Honest limitations

This is synthetic robustness. The synthetic risk signal lives in the vitals and engineered blocks (the project’s own ablation finding), so waveform robustness partly reflects the encoders and partly that the embeddings matter little on synthetic data. The quality gates were bypassed on purpose, so these numbers do not measure the gates themselves. With 2 replay patients the numbers are point estimates. Real data robustness of the embedding path is bounded by the Sudden Cardiac Death Holter Database result, where the PPG stream is entirely absent and no variant beats chance at 6 hours. Nothing in this document is clinical evidence.

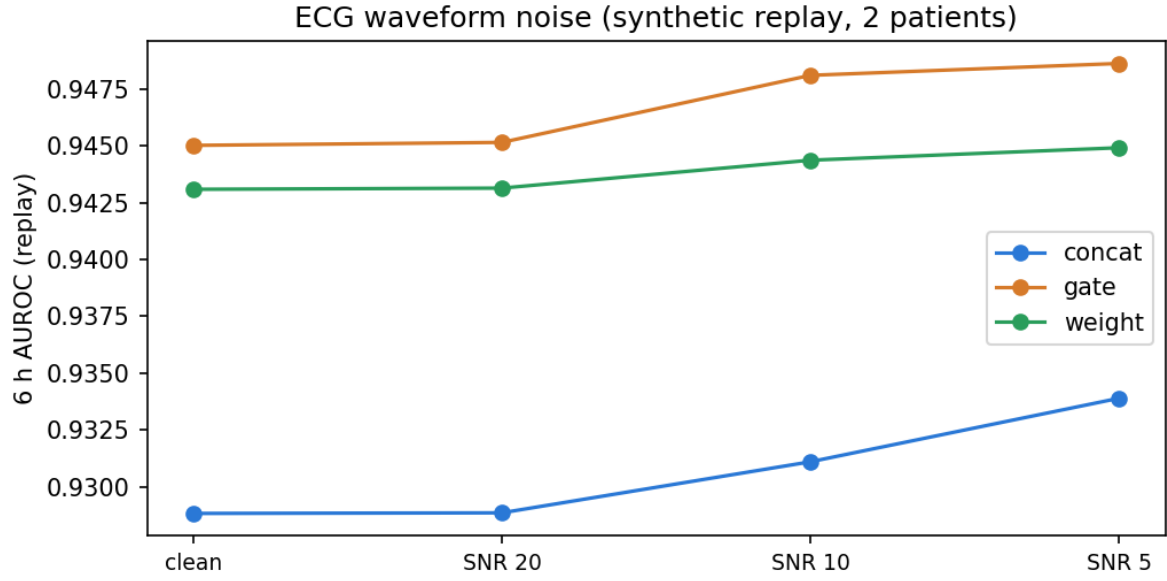


Figure 1: 6 h AUROC versus ECG waveform noise level (synthetic replay). No degradation down to SNR 5 dB.

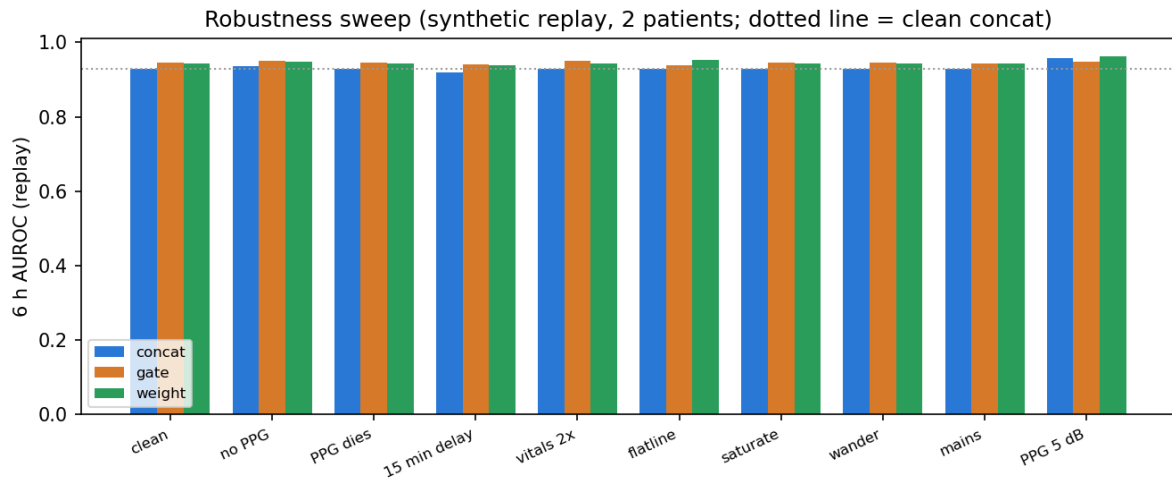


Figure 2: Robustness sweep across all perturbation families (synthetic replay; the dotted line is the clean concat score).